

27 September 2024

Agreement

Concerning the Adoption of Harmonized Technical United Nations Regulations for Wheeled Vehicles, Equipment and Parts which can be Fitted and/or be Used on Wheeled Vehicles and the Conditions for Reciprocal Recognition of Approvals Granted on the Basis of these United Nations Regulations*

(Revision 3, including the amendments which entered into force on 14 September 2017)

Addendum 134 – UN Regulation No. 135

Revision 2 - Amendment 2

Supplement 2 to the 02 series of amendments – Date of entry into force: 15 June 2024

Uniform provisions concerning the approval of vehicles with regard to their Pole Side Impact performance (PSI)

This document is meant purely as documentation tool. The authentic and legal binding text is: ECE/TRANS/WP.29/2023/123 and Corr.1.



UNITED NATIONS

* Former titles of the Agreement:

Agreement concerning the Adoption of Uniform Conditions of Approval and Reciprocal Recognition of Approval for Motor Vehicle Equipment and Parts, done at Geneva on 20 March 1958 (original version); Agreement concerning the Adoption of Uniform Technical Prescriptions for Wheeled Vehicles, Equipment and Parts which can be Fitted and/or be Used on Wheeled Vehicles and the Conditions for Reciprocal Recognition of Approvals Granted on the Basis of these Prescriptions, done at Geneva on 5 October 1995 (Revision 2).



Paragraphs 2.3. and 2.4., amend to read:

- "2.3. *"Compressed hydrogen storage system (CHSS)"* means a system designed to store compressed hydrogen fuel for a hydrogen-fuelled vehicle and composed of a container, container attachments (if any), and all primary closure devices required to isolate the stored hydrogen from the remainder of the fuel system and the environment.
- 2.4. *"Container (for hydrogen storage)"* means the pressure-bearing component on the vehicle that stores the primary volume of hydrogen fuel in a single chamber or in multiple permanently interconnected chambers."

Insert new paragraph 2.4.1., to read

- "2.4.1. *"Container Attachments"* mean non-pressure bearing parts attached to the container that provide additional support and/or protection to the container and that may be only temporarily removed for maintenance and/or inspection only with the use of tools."

Paragraph 2.10., amend to read:

- "2.10. *"Hydrogen-fuelled vehicle"* means any motor vehicle that uses compressed gaseous hydrogen as a fuel to propel the vehicle, including fuel cell and internal combustion engine vehicles. Hydrogen fuel for the vehicles is specified in ISO 14687:2019 and SAE J2719_202003."

Paragraph 2.14., amend to read:

- "2.14. Passenger compartment
- 2.14.1 *"Passenger compartment with regard to occupant protection"* means the space for occupant accommodation, bounded by the roof, floor, side walls, doors, outside glazing, and front bulkhead and the plane of the rear compartment bulkhead or the plane of the rear-seat back support.
- 2.14.2. *"Passenger compartment for electric safety and/or hydrogen safety assessment"* means the space for occupant accommodation, bounded by the roof, floor, side walls, doors, outside glazing, front bulkhead and rear bulkhead, or back door, as well as by the electrical protection barriers and enclosures provided for protecting the occupants from direct contact with high voltage live parts. "

Paragraph 2.15., shall be deleted

Paragraphs 2.16. to 2.20.(former), renumber as paragraphs 2.15. to 2.19.

Paragraph 2.21 (former), renumber as paragraph 2.20 and amend to read:

- "2.20. *"Shut-off valve (for hydrogen-fuelled vehicles)"* means a valve between the container and the vehicle fuel system that must default to the "closed" position when not connected to a power source. "

Paragraphs 2.22. to 2.26.(former), renumber as paragraphs 2.21. to 2.25.

Paragraph 2.27 (former), renumber as paragraph 2.26 and amend to read:

- "2.26. *"Vehicle type"* means a category of vehicles, the design characteristics of which do not differ, in so far as they have an adverse effect on the result of the impact test prescribed in this Regulation, in such essential respects as:
- (a) The type of protective system(s);
 - (b) The type of front seat(s);
 - (c) The vehicle width;
 - (d) The wheelbase and overall length of the vehicle;
 - (e) The structure, dimensions, lines and materials of the side walls of the passenger compartment, including any optional arrangements or

interior fittings within or about the side walls of the passenger compartment;

- (f) The type of door latches and hinges;
- (g) The type of fuel system(s);
- (h) The unladen vehicle mass and the rated cargo and luggage mass;
- (i) The sitting of the engine (front, rear or centre);
- (j) The locations of the REESS."

Paragraphs 2.28. to 2.56.(former), renumber as paragraphs 2.27. to 2.55.

Annex 6, paragraph 2.1., amend to read:

"2.1. "Enclosed spaces" means the special volumes within the vehicle (or the vehicle outline across openings) that are external to the hydrogen system (storage system, fuel cell system, internal combustion engine (ICE) and fuel flow management system)."

Annex 6, paragraph 3.1.4., amend to read:

"3.1.4. The main stop valve and shut-off valves for hydrogen gas, located in the downstream hydrogen gas piping, are in the normal driving condition kept open immediately prior to the impact."

Annex 6, paragraphs 4.2. and 4.3., amend to read:

"4.2. The initial mass of hydrogen in the storage system can be calculated as follows:

$$P_o' = P_o \times 288 / (273 + T_o)$$

$$\rho_o' = -0.0027 \times (P_o')^2 + 0.75 \times P_o' + 1.07$$

$$M_o = \rho_o' \times V_{CHSS}$$

4.3. Correspondingly, the final mass of hydrogen in the storage system, M_f , at the end of the time interval, Δt , can be calculated as follows:

$$P_f' = P_f \times 288 / (273 + T_f)$$

$$\rho_f' = -0.0027 \times (P_f')^2 + 0.75 \times P_f' + 1.07$$

$$M_f = \rho_f' \times V_{CHSS}$$

where P_f is the measured final pressure (MPa) at the end of the time interval, and T_f is the measured final temperature (°C)."
